

Please write clearly in block capitals.

Centre number

Candidate number

Surname

Forename(s)

A-level

ENVIRONMENTAL SCIENCE**Paper 1 (Generated practice paper - Set 19)**

Assessed: The physical environment, Energy resources, Pollution, Research methods

Time allowed: 3 hours**Materials**

For this paper you may use:

- a calculator.

Instructions

- Use black ink or black ball-point pen. Pencil should only be used for drawing.
- Fill in the boxes at the top of this page.
- Answer **all** questions 1 to 10 and **one** essay from question 11.
- You must answer the questions in the spaces provided. Do not write outside the box around each page or on blank pages.
- Do all rough work in this book. Cross through any work you do not want to be marked.

Information

- The marks for questions are shown in brackets.
- The maximum mark for this paper is 120.
- All questions should be answered in continuous prose.
- You will be assessed on your ability to:
 - use good English
 - organise information clearly
 - use specialist vocabulary where appropriate.

For Examiner's Use

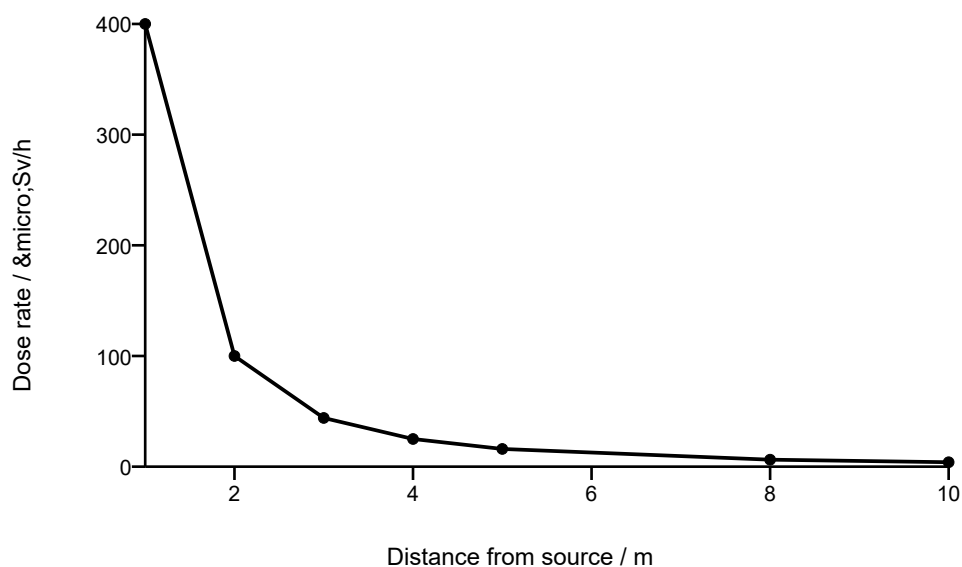
Question	Mark
1	
2	
3	
4	
5	
6	
7	
8	
9	
10	
11	
TOTAL	

Answer all questions in the spaces provided.

0 1

Figure 1 shows the radiation dose rate measured at different distances from a small gamma source.

Figure 1



0 1

1

Use Figure 1 to show that the dose rate follows the inverse square law.

[2 marks]

.....

.....

.....

.....

0 1

2

Calculate the dose rate at 20 m from the source.

[1 mark]

Answer

$\mu\text{Sv/h}$

0 1

3

Explain how each of the following reduces the exposure of workers handling radioactive sources.

[4 marks]

Closed (sealed) sources

.....

Absorbing materials

.....

Reduced exposure time

.....

Worker monitoring

.....

0 1 . 4

Explain why potassium iodide tablets are issued to people near a nuclear accident.

[2 marks]

.....

.....

.....

0 1 . 5

State the meaning of ALARA.

[1 mark]

.....

.....

10

0 2 . 1

Which of the following best describes the purpose of Critical Group Monitoring? Tick one box.

[1 mark]

A To detect health damage in the population most affected by a pollutant ☒

B To predict the route a pollutant will take through the environment ☐

C To assess exposure of the people most at risk so that everyone else ☒ can be assumed to be safe

D To measure the concentration of a pollutant at the point of release ☐

0 2 . 2

Explain why CPA can reduce the cost of an environmental monitoring programme.

[2 marks]

.....

0 2 . 3

A new chemical has properties similar to PCBs (persistent, liposoluble chlorinated organic compounds). Explain how knowledge of PCBs helps decisions about the new chemical.

[2 marks]

5

0 3

Table 1 shows the water quality required for four different uses.

0 3 . 1

Complete Table 1 by giving the quality requirement for each water use and the reason for it.

[4 marks]

Table 1

Water use	Quality requirement	Reason	
Potable (public) supply			
Spray irrigation			
Power station condenser cooling			
Industrial boiler water			

0 3 . 2

Explain why aeration is used in the treatment of water from anaerobic sources.

[2 marks]

0 3 . 3

Explain how flocculation removes clay particles that sedimentation cannot.

[2 marks]

0 3 . 4

Give **two** methods used to sterilise treated water.

[2 marks]

1

2

10

0 4 . 1

Explain how integrated manufacture saves energy in the iron and steel industry.

[2 marks]

0 4 . 2

Explain why recycling is not always energy efficient.

[2 marks]

0 4 . 3

Explain how each of the following manages electricity infrastructure to conserve energy.

[4 marks]

High-voltage grid

.....

.....

.....

ICT management of supply and demand

.....

.....

.....

0 4 . 4

Give **two** locational factors affecting where new generating infrastructure is built.

[2 marks]

1

.....

2

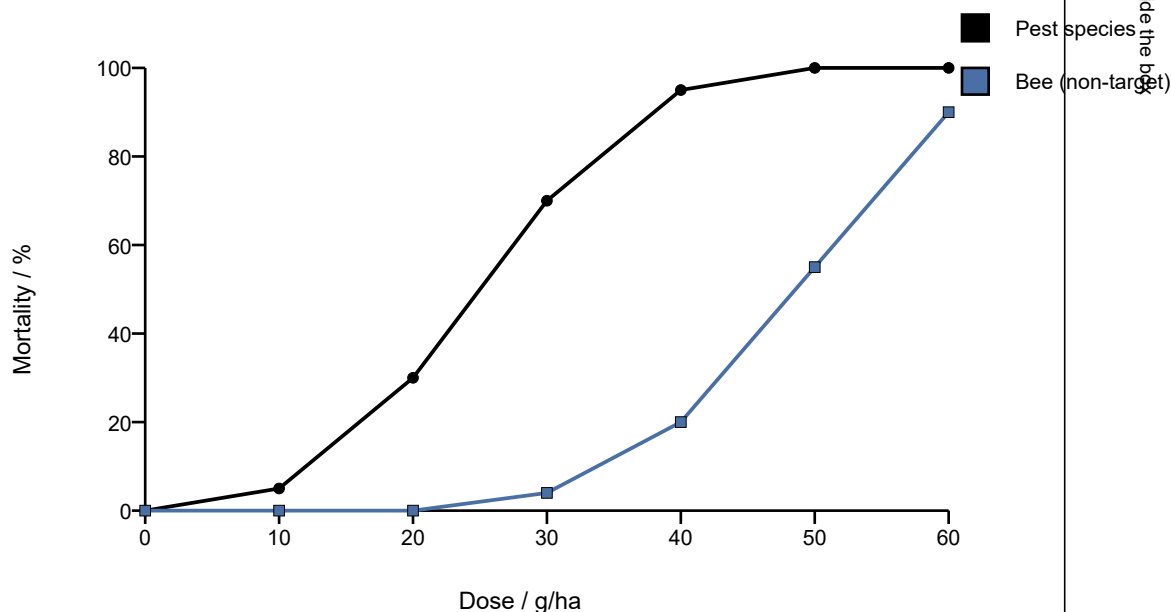
.....

10

0 5

Figure 1 shows how the percentage mortality of a pest species and of a non-target bee species change with the dose of an insecticide applied to a crop.

Figure 1



0 5

1

Use Figure 1 to estimate the LD_{50} of the insecticide for the pest species.

[1 mark]

Answer

 g/ha

0 5

2

Define the term **MDNF** (maximum dose never fatal).

[1 mark]

0 5

3

Use Figure 1 to explain why it may not be possible to kill all of the pest species without harming bees.

[3 marks]

0 5 . 4

Suggest how the farmer could reduce the harm to bees without reducing the dose applied.

[2 marks]

0 5 . 5

Explain why a pesticide with high specificity is less likely to cause environmental damage.

[3 marks]

10

0 6 . 1

Describe how an **electrostatic precipitator** removes smoke particles from the effluent gases of a coal-fired power station.

[3 marks]

0 6 . 2

Explain how a **cyclone separator** works.

[2 marks]

.....

.....

.....

0 6 . 3

Explain why heating coal to remove tar reduces smoke emissions.

[1 mark]

.....

0 6 . 4

Explain why turbochargers reduce the smoke produced by diesel engines.

[2 marks]

.....

.....

.....

0 6 . 5

Explain why smaller smoke particles (PM1) are a greater health risk than larger ones.

[2 marks]

.....

.....

.....

10

07

A student estimated the population of ground beetles in a field using the Lincoln Index. 60 beetles were caught in pitfall traps, marked with a spot of paint and released. Three days later 45 beetles were caught, of which 9 were marked.

07

. 1

Calculate the estimated population size.

[2 marks]

Answer

beetles

07

. 2

State **four** assumptions of the Lincoln Index method.

[4 marks]

1

.....

2

.....

3

.....

4

.....

07

. 3

Explain how the estimate would be affected if the paint made the beetles more likely to be eaten by birds.

[2 marks]

.....

.....

.....

.....

07

. 4

Suggest why the second sample was taken after three days rather than the next day or after a month.

[2 marks]

.....

.....

10

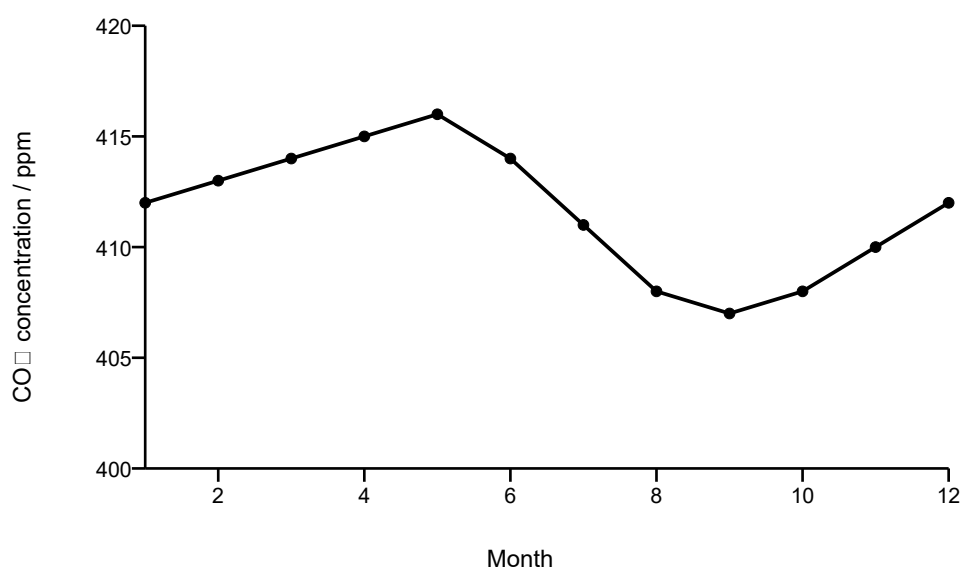
08

Table 1 shows the composition of the dry atmosphere. Figure 1 shows the changes in carbon dioxide concentration measured at a site in the northern hemisphere over one year.

Table 1

Gas	Percentage by volume / %
Nitrogen	78
Oxygen	21
Argon and other rare gases	0.9
Carbon dioxide	0.04
Ozone	0.000007

Figure 1



08

1

Describe and explain the pattern shown in Figure 1.

[4 marks]

08 . 2

Explain why the atmosphere is described as being in a state of **dynamic equilibrium**.

[2 marks]

08 . 3

Give **two** ways in which the atmosphere supports life other than by providing gases for biological molecules.

[2 marks]

1

2

08 . 4

Name the two layers of the atmosphere most affected by human activities.

[2 marks]

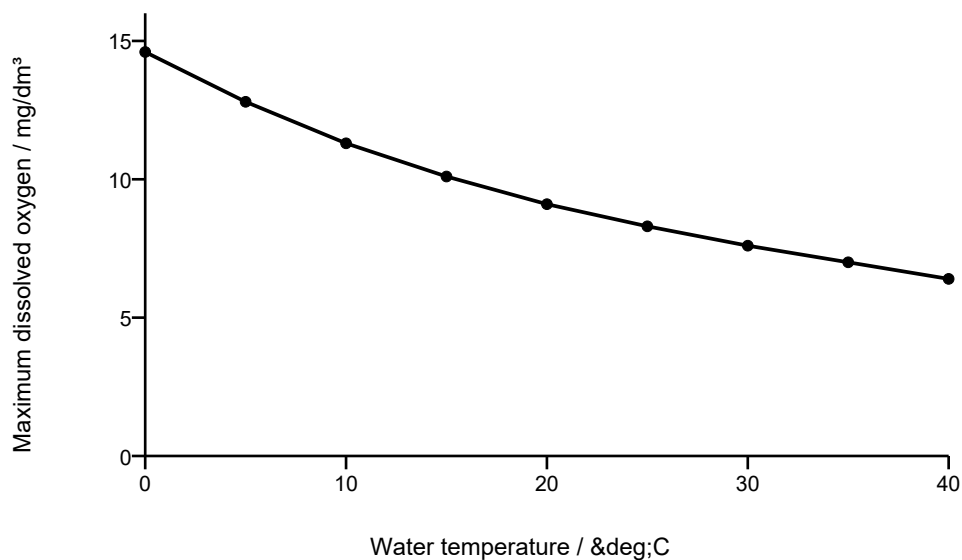
1

2

10

09

Figure 1 shows the maximum concentration of dissolved oxygen in fresh water at different temperatures. A power station discharges warm cooling water into a river. The lowest dissolved oxygen concentration that trout can tolerate is 6.5 mg dm^{-3} .

Figure 1**09****1**

Describe the relationship shown in Figure 1.

[1 mark]**09****2**

Use Figure 1 to determine the highest water temperature at which trout could survive if the water were fully saturated with oxygen.

[1 mark]

Answer

&deg;C

09**3**

In summer the river upstream of the power station is at 20°C and the effluent raises the temperature by 7°C . Use Figure 1 to calculate the reduction in maximum dissolved oxygen concentration caused by the effluent.

[2 marks]

Answer mg/dm³

09 . 4

Explain why the actual dissolved oxygen concentration downstream may be much lower than the values in Figure 1.

[2 marks]

.....

.....

.....

.....

09 . 5

Describe **three** effects of warmer water on aquatic organisms other than reduced dissolved oxygen.

[3 marks]

1

.....

2

.....

3

.....

09 . 6

Describe how a cooling tower reduces the temperature of power station effluent water.

[1 mark]

.....

.....

10

Explain why the development of energy storage technologies is essential if renewable energy resources are to provide most of a country's electricity.

ESP/7447/1/SET19

Extra space

10 . 2 State what is meant by **peak shaving**.

[1 mark]

10

Do not write outside the box

Write an essay on one of the following topics.

1	1	.	1
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Explain how feedback mechanisms and tipping points influence the rate of global climate change and discuss the implications for the strategies needed to control it.

[25 marks]

OR

1 1 . 2

Discuss how knowledge of the properties of energy resources can be used to reduce the severity of their impact on the environment.

[25 marks]

Indicate which question you have answered by writing its number, **11 . 1** or **11 . 2**, at the top of your answer.

[illegible]

[illegible]

[illegible]

END OF QUESTIONS